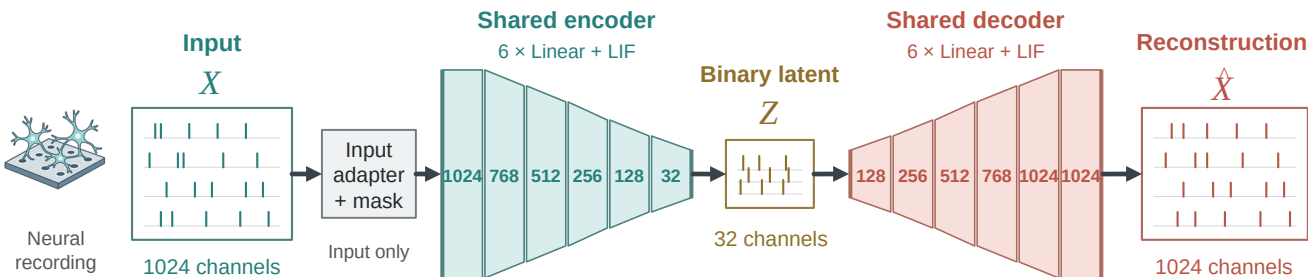


## (a) Spike-to-spike compression

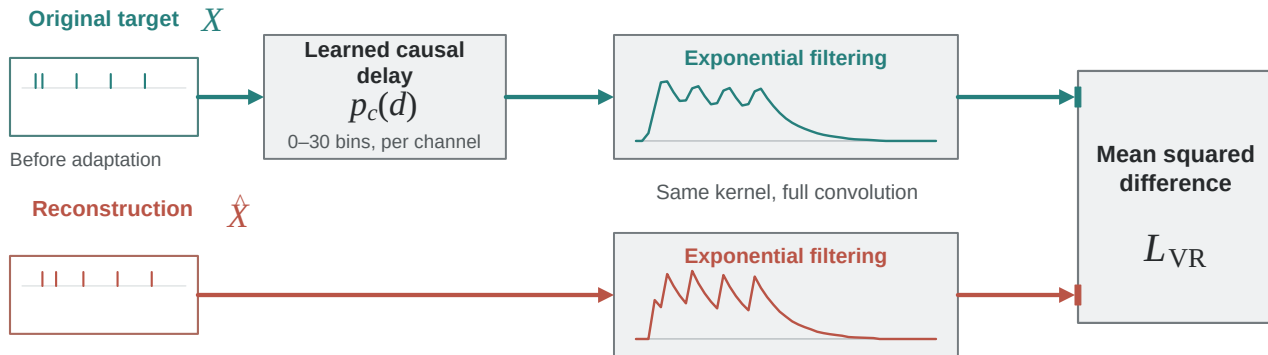


Encoder: 1024  $\rightarrow$  1024  $\rightarrow$  768  $\rightarrow$  512  $\rightarrow$  256  $\rightarrow$  128  $\rightarrow$  32

Decoder: 32  $\rightarrow$  128  $\rightarrow$  256  $\rightarrow$  512  $\rightarrow$  768  $\rightarrow$  1024  $\rightarrow$  1024

Time runs left to right and is preserved through the shared stages.

## (b) Delay-tolerant training



### Illustrative causal delay

One event spreads to later bins



Schematic spikes and widths. Training input and target include 30 zero-tail bins.